

```
clear; clc; close all;
```

```
% Data loading and formatting
```

```
data = readtable('Verlorenvlei_Groundwater_isotopes.xlsx', ...  
    'VariableNamingRule','preserve');
```

```
data.Date = datetime(data.Date);
```

```
m = month(data.Date);  
yrs = year(data.Date);
```

```
grp = strings(size(data.Date));
```

```
% Creating legends according to date of collection
```

```
grp(m == 3 & yrs == 2023) = "Mar 2023";  
grp(m == 6 & yrs == 2023) = "Jun 2023";  
grp(m == 11 & yrs == 2023) = "Nov 2023";  
grp(m == 1 & yrs == 2024) = "Jan 2024";
```

```
d2H = data{:, 'δ2H (‰)'};  
d180 = data{:, 'δ180 (‰)'};
```

```
% Data validation
```

```
valid = ~isnan(d2H) & ~isnan(d180);
```

```
d2H = d2H(valid);  
d180 = d180(valid);  
grp = grp(valid);
```

```
% Plot labelling and color
```

```
lbl = ["Mar 2023", "Jun 2023", "Nov 2023", "Jan 2024"];
```

```
clrs = [  
    0.0 0.4 1.0;  
    0.0 0.7 0.7;  
    0.0 0.7 0.2;  
    1.0 0.2 0.2;  
];
```

```
x = linspace(min(d180)-1, max(d180)+1, 100);  
GMWL = 8*x + 10;
```

```
figure; hold on;
```

```
h = gobjects(5,1);
```

```
% For loop plot
```

```
for i = 1:4  
    idx = grp == lbl(i);
```

```

h(i) = scatter(d180(idx), d2H(idx), 80, ...
    'filled', 'MarkerFaceColor', clrs(i,:));
end

h(5) = plot(x, GMWL, 'k:', 'LineWidth', 2);

% Axes labelling
xlabel('\delta^{18}O (‰)');
ylabel('\delta^{2}H (‰)');
title('Verlorenvlei Groundwater Stable Isotopic Trend');

grid on;
box on;

legend(h, [lbl, "GMWL"], 'Location', 'best');

hold off;

```

